

IAF HUMAN SPACEFLIGHT SYMPOSIUM (B3)
Human Spaceflight Global Technical Session (9-GTS.2)

Astronaut Profile Evolution study: Analyzing Evolution Since 1961 - How Has Society Shaped the Ideal Astronaut?

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Abstract

Since Yuri Gagarin's historic flight in 1961, the selection criteria for astronauts have undergone significant evolution, shaped by societal changes, technological advancements, and shifting exploration objectives. The Diversity and Gender Equality Project Group of the Space Generation Advisory Council initiated the DIVERsity IN Astronaut Selection (DIVINAS) project to scrutinize this evolution. This paper presents the findings of the DIVINAS project, which aims to analyze the changing profile of astronauts over time and assess the impact of societal factors on astronaut selection. Through meticulous data collection and comparative analysis, this study examines key parameters and capacities required of astronauts across different eras, from the pioneering space missions to contemporary crew selections. It investigates recent recruitments in both private and public space agencies, considering factors such as family life, cultural diversity, and the inclusion of individuals with disabilities in astronaut programs.

By correlating historical data with societal trends, the study seeks to delineate the ideal astronaut profile, taking into account regional variations and temporal shifts. Moreover, it endeavors to anticipate the characteristics that will define successful astronauts in the coming years, as humanity gears up for ambitious exploration endeavors beyond Earth orbit.

The findings of this research not only shed light on the evolving nature of astronaut selection but also provide valuable insights for future space exploration initiatives. Understanding what makes a great astronaut in different contexts is crucial for shaping inclusive and effective space exploration programs that harness the diverse talents and capabilities of humanity.

This study contributes to the ongoing dialogue on the future of space exploration and the role of astronauts as ambassadors of humanity's collective aspirations beyond the confines of our home planet. Through a comprehensive examination of astronaut profiles and the factors influencing their selection, this research provides a roadmap for optimizing astronaut recruitment processes, fostering inclusivity, and maximizing the potential for successful space missions in the decades to come.

Keywords: astronaut selection, space missions, crew size, training programs, diversity in space, database.

Acronyms/Abbreviations

DIVINAS: Diversity in Astronaut Selection
ESA: European Space Agency
ISS: International Space Station
LEO: Low Earth Orbit
NASA: National Aeronautics and Space Administration
ALCE: Latin American and Caribbean Space Agency
VR: Virtual Reality
AR: Augmented Reality
SFU: Space Flyer Unit
STS: Space Transportation System (Space Shuttle Program)
EVA: Extra-Vehicular Activity

1. Introduction

1.1 Background and Purpose

The process of selecting astronauts has always been crucial to the success of space missions, evolving alongside advancement in space exploration. Initially, the selection criteria were narrowly focused on physical fitness, military experience, and technical expertise. However, as space missions have become more diverse and complex, so too have the requirements for astronauts. The rise of private space companies, international collaborations, and the increasing importance of diversity and inclusivity have broadened the criteria for astronaut selection.

1.2 Previous Work

The DIVINAS (Diversity in Astronaut Selection) research team has previously explored various facets of astronaut selection, including gender diversity, ethnic representation, and the professional background of selected astronauts. These studies have highlighted the significance of inclusivity and the need to adapt selection criteria to reflect the changing demands of the space sector. However, there remain areas that have not been fully explored, such as regional trends, the impact of technological advancements, and the evolution of training programs.

1.3 Objectives for the Current Study

Building on the foundation of previous work, this study aims to explore new dimensions of astronaut selection that have not been extensively covered before. Specifically, the paper will analyze regional trends and

national diversity, examine the evolution of crew sizes and mission types, and investigate changes in astronaut training programs. Additionally, the study will assess the influence of societal changes and technological advancements on astronaut selection criteria. Through this expanded analysis, the paper seeks to provide a comprehensive understanding of how these factors have shaped the profiles of astronauts over time and to contribute to the development of a more inclusive and effective selection process.

2. Astronaut Database: Leveraging Existing Data for New Insights

2.1 Dataset Description

Existing data from the past launches is useful for making precise statistics and drafting conclusions on specific topics. However, it is also interesting to mix it with information from current astronauts to proof check and also to get another dimension to the data.

The dataset starts in 1961. This includes publicly-funded astronauts, cosmonauts, and taikonauts hired by government agencies as well as people on privately funded missions. A spreadsheet was created with a row dedicated to each crew member per spaceflight, with people who have flown to space multiple times having one row per launch.

The following information was recorded about each person at the time of launch, where available:

1. Name
2. Gender
3. Nationality
4. Second nationality (if applicable)
5. Ethnicity
6. Second ethnicity (if applicable)
7. Number of years of relevant experience
8. Occupation prior to selection
9. Sport played
10. Second sport played
11. Whether they have served in military
12. Whether they had pilot training prior to selection
13. Whether they had scuba diving training prior to selection
14. Space agency they belong to or Private
15. Crew size for the launch

16. Launch date
17. Launching country and whether the launcher is private or public
18. Mission type
19. Second mission type (if applicable)
20. Mission duration

2.2 Grouping of Answers

To simplify data collection and to enable comparison of statistics, possible answers for certain fields were grouped. Ethnicity groups were: American Indian or Alaska Native, Asian, Black or African American, Hispanic or Latino, Native Hawaiian or Other Pacific Islander and White, in accordance with the United State Office of Management and Budget 1997 Standards for Maintaining, Collecting, and Presenting Federal Data on Race and Ethnicity [1]. Occupations were also grouped into categories such as medicine, pilot, several engineering disciplines, several science disciplines, and others. Military experience, flying experience, and diving experience could be filled in with “Yes”, “No”, or “N/A” if information was not available. Launching country and type could be filled in with “American governmental”, “American private”, “Russian government”, or “Chinese governmental”. NASA’s recent missions on SpaceX’s Crew Dragon, for example, were labelled “American governmental” as they were launched from the United States but operated by a commercial provider. Mission types were grouped into: Firsts (early tests), satellite deployment, scientific experiment, station maintenance or deployment, commercial or advertisement, tracking and data relay satellite, earth observation, arts, lunar surface operations, telescope or spacelab deployment, telescope maintenance, and cargo resupply.

3. Regional Trends and National Diversity in Astronaut Selection

3.1 Analysis of Regional Representation Over Time

The shift in astronaut representation from 1961 to 2024 underscores a notable transformation in the landscape of space exploration. Initially dominated by astronauts from the United States and Russia, the scope has significantly expanded to include nations previously absent from space endeavors.

Below is an image that shows worrying numbers regarding the number of astronauts compared to nations that already dominated the space scene, for example the United States with 60% of astronauts worldwide, followed by Russia and the Soviet Union, which together account for around 20%, followed by China with 3.30% representation.

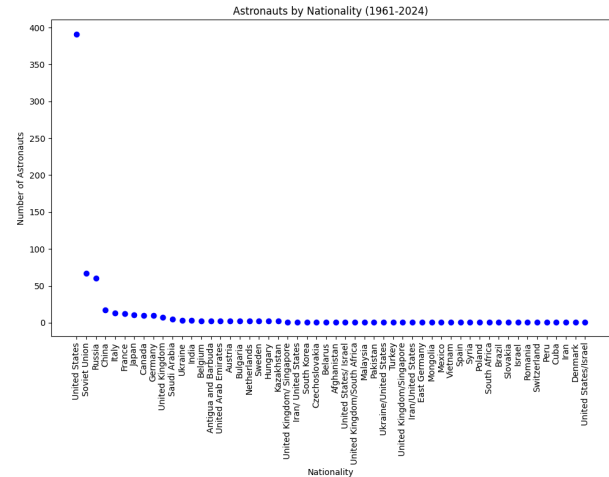


Figure 3-1: Astronauts by nationality trend over decades.

The nations of the European Union, Japan and Saudi Arabia are countries that were not usually in the space race, but are now very present.

This evolution is partly fueled by the advent of civilian-only spaceflights, facilitating access for countries without traditional space programs. Such inclusivity heralds a new era of global participation in space, democratizing access and reflecting broader societal advancements in technology and international cooperation in the field of astronautics.

3.2 Examining National Diversity Among Astronauts

Talking about civilian-only spaceflight, suborbital and even orbital, there are 138 new spaceflight entrants from 2021 to September 2024. In this study sample, we can see that the majority of the countries most present in space are still the traditional ones in the space race. However, even at a slow pace, we see an unprecedented accessibility with countries without even a space program reaching space, as can be seen in the following graph.

While nations like the United States and Russia dominate astronautical contributions, others such as India, despite its space achievements, have fewer astronauts. It also highlights the socio-economic challenges faced by rapidly developing countries like Bangladesh, Sri Lanka, and Nepal, which hinder their space presence. Additionally, the study explores the participation of Latin American countries, particularly the establishment of the Latin American and Caribbean Space Agency (ALCE) in 2021, reflecting regional efforts to enhance space capabilities (Gres, et al., 2023).

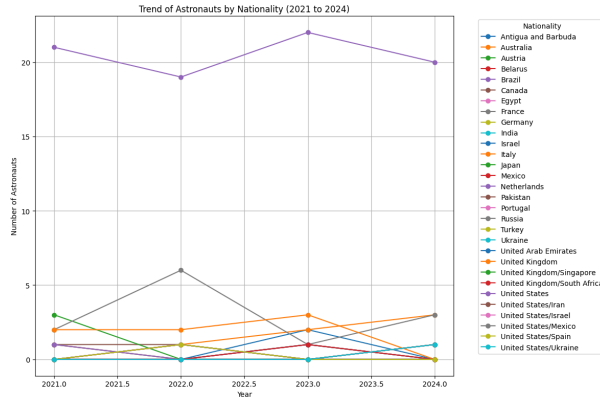


Figure 3-2: Civil astronauts by nationality (2021-2024)

3.3 Qualitative Analyses of Regional and National Trends

In examining the evolving profile of astronauts, Victor Hespanha's personal reflections provide valuable qualitative insights that resonate with the quantitative analyses presented below. Hespanha, selected as an astronaut in 2022 through Blue Origin, underscores the significance of cultural background over the specific space program in shaping his long-term aspirations to venture into space. This perspective aligns with Gres et al.'s (2022)(2023) findings, which highlight the increasing diversity of nationalities among astronauts as space agencies expand their international collaborations and inclusivity efforts.

Hespanha's experience as a Brazilian astronaut, emerging from a non-traditional background in the financial sector, exemplifies the paper's assertion that the criteria for astronaut selection have broadened beyond the historical predominance of military pilots and engineers. His selection process, characterized by accessibility and non-traditional pathways, mirrors the paper's discussion on the diversification of professional backgrounds in recent decades. Furthermore, Hespanha's observation that societal changes influence the ideal astronaut profile is corroborated by Gres et al. (2022), who document how evolving societal values and technological advancements have necessitated a more inclusive and varied astronaut demographic.

Additionally, Hespanha's brief mission aboard the New Shepard vehicle, involving a six-person crew and a highly organized training regimen, reflects the paper's analysis of contemporary spaceflight missions that emphasize teamwork and adaptability within increasingly diverse and multinational crews. His acknowledgment of the rapid development curve and the importance of networking for future astronauts echoes the paper's recommendations for fostering diversity and preparing for the dynamic demands of modern space exploration.

4. Crew Size and Mission Type: Correlations and Evolution

4.1 Evolution of Crew Sizes Across Mission Types

Over the history of human spaceflight, different variations of crew sizes can be observed over the timeline. Plethora of studies can be found in the literature, investigating various factors that contribute in deciding a human crew organisation for a mission and attempting to find the optimum crew size that would suit for various mission objectives. Through analogue missions, practical experiments have been conducted to understand various aspects and dependencies that occur in a human space mission. A range of crew sizes can be also seen in these simulated space environments; for example, the BIOS-3 experimental closed ecosystem [4-1], was used to conduct one to three manned crew closure experiments in the 1970s and 1980s, with the longest experiment being a three-man crew which lasted 180 days. MARS-500 [4-2] is another example of an isolation experiment which was intended to study the psychological, physiological, and technological challenges inherent to long-duration space flight. It spanned between 2007 and 2011 and had a maximum capacity to accommodate a crew of six. Among various factors of a space mission, the specifications of the structure housing the astronauts at any time in a space mission, drives the maximum limit on the crew size; whether it's the launch vehicle, the space station or habitable modules on another planet or simulated facilities on Earth. The mandatory technical personnel needed to operate this housing structure and the complexity of a mission type, drives the minimum requirement on the crew size. For example, the SpaceX's Crew Dragon spacecraft normally carries a crew of four [4-3], but was originally designed to carry seven people and needs a minimum of two people (a commander and a pilot) to fly its spacecraft.

From Figure 4-1 and Figure 4-2 it can be observed that a crew size of 3 astronauts has been the most popular number for a space mission throughout human space history; mainly dominating the late sixties-early seventies and the decade of 2011-2020 as compared to missions with other crew sizes. A crew size of 2 and 7 astronauts cover the next highest proportion, whereas a crew of 8 and 4 astronauts being the least favoured for a space mission - the only space mission with a crew of 8 in the history of human spaceflight being the STS-61-A that was launched on the Space Shuttle Challenger in 1985. From Figure 4-2, it is also observed that missions with crew size 7 covered the years from 1980s to 2011, which corresponds to the Space Shuttle program's

tenure that had a larger carrying capacity of up to 8 astronauts.

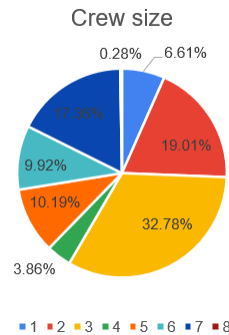


Figure 4-1: Crew size distribution for space missions.

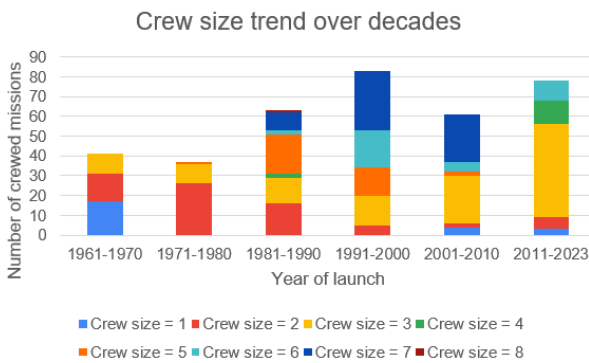


Figure 4-2: Crew size trend over decades.

As covered in section 2 above, the database includes the type of each mission as two main categories - Mission Type 1 being the primary objective of a mission and Type 2 being a secondary (optional) objective. Each of these types are further classified into 12 different sub-categories. Figure 4-3 shows, largest proportion of all the space missions are conducted with the primary objective of Station maintenance or deployment and to perform scientific experiments. The only mission launched with its main objective of retrieving another satellite/spacecraft was the STS-72 capture and return to Earth mission of a Japanese microgravity research spacecraft known as Space Flyer Unit (SFU) in 1996. The 'unknown' category is included to cover cases in the database where data on the mission type was not readily available, or there was no secondary objective in the case of Mission type 2 (secondary mission objective) category.

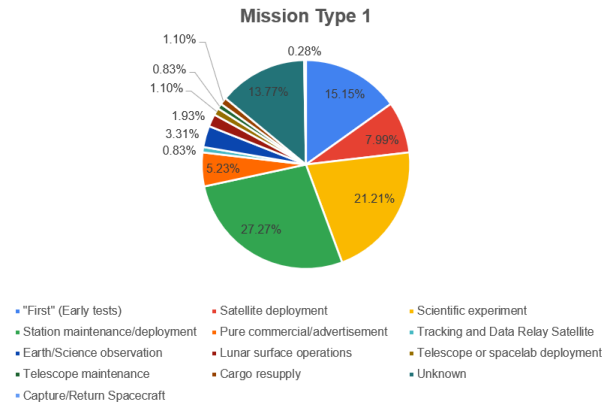


Figure 4-3: Mission Type 1 distribution for space missions.

In order to effectively illustrate the whole timeline of different mission types, a reduced set of categories have been considered - with mission types of Satellite deployment, Station maintenance/deployment, Tracking and Data Relay Satellite, Telescope or spacelab deployment and Telescope maintenance grouped into one category called 'System deployment / maintenance' and mission types of Scientific experiment and Earth/Science observation grouped into one category called 'Science / EO', as shown in Figure 4-4.

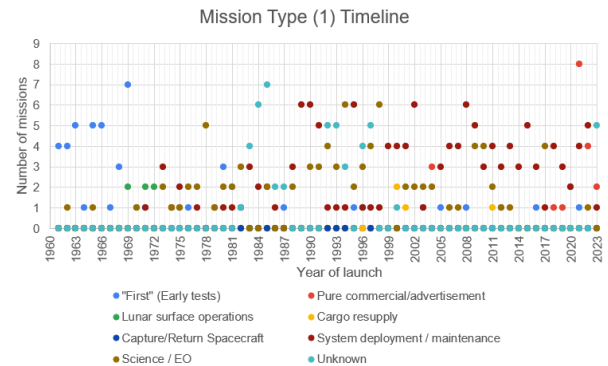


Figure 4-4: Mission Type 1 timeline.

4.2 Analyzing Mission Types and Their Impact on Crew Dynamics

The missions in the first 2 decades of human spaceflight were mainly motivated by the space race between the USA and USSR, when humans first started testing the space limits and eventually sent missions to the moon. A smaller crew size of 1, 2 and 3 astronauts were considered favourable for these missions, as observed in Figure 4-4 and Figure 4-5. One of the main launch vehicles used to send astronauts to LEO is the Soyuz spacecraft which has a carrying capacity of 3 people and has been in service since the 1960s.

As the Apollo missions to the Moon came to an end in the early 1970s, focus started shifting on conducting science and research in space and building infrastructure in LEO for long-duration human spaceflights. As the missions started to become more diverse and complex, for example, with the launch of the first ISS module in 1998 and the Hubble Space Telescope servicing missions, requiring astronauts to conduct EVAs and system maintenance and deployment; variations in crew sizes started appearing as well, leaning towards crew sizes of 3 to 7 from 1980s onwards. As the number of launches and human spaceflights started becoming more frequent and accessible, mission objectives became diverse as well. Along with system maintenance, a large proportion of the missions were launched to conduct various scientific experiments. As the ISS became operational, missions dedicated to resupply cargo to the station were also launched in the late 1990s. With the advent of commercialisation of space and emergence of private launch vehicles, missions dedicated for space tourism and citizen science experiments have started to become more frequent in the last decade.

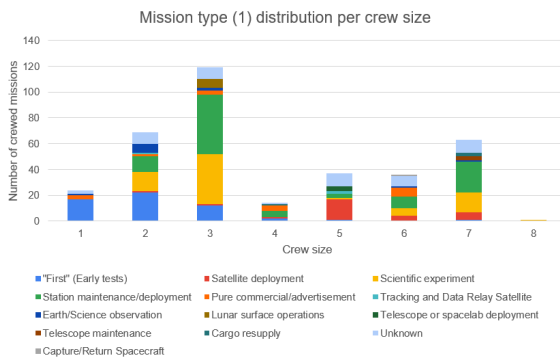


Figure 4-5: Mission type 1 distribution per crew size.

Although, overall, gender diversity seems to be improving gradually over time, females still account for nearly only 14% throughout human spaceflight history, compared to their male counterparts. Combining all missions with their primary objectives of system maintenance/deployment and science/Earth observation, the missions comprised 13.83% of female crew members. Interestingly, the highest participation of female astronauts was observed for mission type pure commercial/advertisement (30% compared to male crew members). This also correlates to increased female participation in general in the recent decade, as can be observed in Figure 6-1.

Gender distribution over Mission type 1

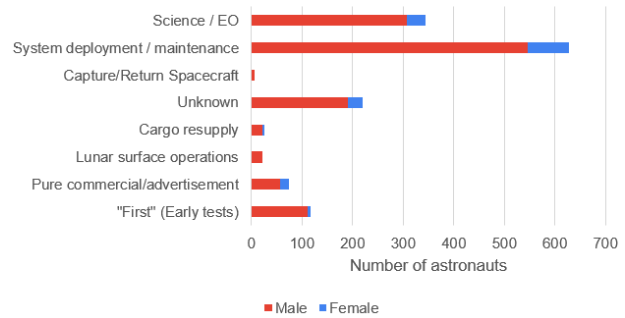


Figure 4-6: Gender distribution over Mission type 1.

An attempt was also made to understand any correlation between primary and secondary objectives of a human space mission. However, no obvious trend or conclusions could be inferred.

		MISSION TYPE 2									
		"First" (Early tests)	Satellite deployment	Scientific experiment	Station maintenance/deployment	Pure commercial/advertisement	Earth/Science observation	Arts	Cargo resupply	Unknown	
M	"First" (Early tests)	0	0	13	3	0	5	1	0	33	
	Satellite deployment	1	0	10	0	0	3	0	0	15	
	Scientific experiment	1	3	0	15	3	10	1	1	43	
	Station maintenance/deployment	1	1	31	0	2	0	0	3	61	
	Pure commercial/advertisement	4	0	0	1	0	0	1	0	13	
	Tracking and Data Relay	0	0	1	1	0	1	0	0	3	
	Satellite	0	0	0	0	0	0	0	0	0	
	Earth/Science observation	1	0	6	0	0	0	0	0	5	
	Lunar surface operations	0	0	0	0	0	5	0	0	2	
	Telescope or spacelab deployment	0	0	0	0	0	3	0	0	1	
E	Telescope maintenance	0	0	0	2	0	0	0	0	1	
	Cargo resupply	0	0	0	3	0	0	0	0	1	
	Unknown	0	0	0	0	0	0	0	0	50	
	Capture/Return Spacecraft	0	0	0	0	0	0	0	0	1	

Figure 4-5: Mission type 1 vs Mission type 2.

5. Astronaut Training Programs: Historical Evolution and Modern Advancements

5.1 Historical Development of Astronaut Training Programs

The historical development of astronaut training programs has evolved dramatically since the early days of space exploration. Initially, astronaut candidates were selected primarily from test pilots, especially during the Mercury, Gemini, and Apollo programs, where the spacecraft had limited maneuvering capabilities, and reentry relied on parachutes. These early astronauts were chosen for their ability to handle high-stress situations rather than their piloting skills, although piloting experience remained a valuable asset.

The introduction of the space shuttle program in 1978 marked a major shift in astronaut training. The shuttle acted both as a spacecraft and a high-speed glider during reentry, necessitating more complex flying skills. Consequently, NASA began selecting two types of astronauts: those with jet aircraft experience, trained to serve as shuttle pilots and commanders, and mission

specialists, chosen for their scientific or technical expertise. This second group focused on operating space station systems and conducting scientific experiments, including spacewalks. The addition of mission specialist roles in the 1970s and 80s, such as educator astronauts and international mission specialists, expanded the diversity of astronaut backgrounds and training.

As space missions grew longer with the advent of the International Space Station (ISS), astronaut training adapted to focus on problem-solving and general spaceflight skills, especially for long-duration missions. Training for astronauts now includes survival skills, spacewalking, handling weightlessness, and mastering complex spacecraft systems. The Russian cosmonaut program also influenced astronaut training, with its emphasis on preparing for extended space flights since the 1990s.

Today, astronaut training programs are global, with contributions from Russia, Europe, and China, and have expanded to include more comprehensive physical and mental preparations, reflecting the growing complexity of modern space missions. [8]

6. Societal and Technological Impacts on Astronaut Selection

6.1 Societal Changes Influencing Selection Criteria

Figure 6-1 illustrates the percentage of female members among the crew for all missions till date. It can be observed that for most of the earlier smaller crew size missions (1 to 3), they were all-men crews. Gender equality movements and global diversity initiatives have significantly influenced astronaut selection processes in recent years. Initiatives specifically aimed at promoting gender equality and diversity have led to a more inclusive selection process. For example, NASA's Artemis program aims to land the first woman and the next man on the Moon, reflecting a commitment to gender diversity. Space agencies worldwide are adopting policies that promote diversity. The European Space Agency (ESA) has implemented measures to ensure a more balanced representation of genders in their astronaut corps. International cooperation in space missions has also fostered diversity. Countries like the United Arab Emirates and China are making strides in including women in their space programs. In general, programs targeting young women and minorities aim to inspire and prepare the next generation of diverse astronauts. These initiatives often include scholarships, mentorship programs, and STEM education campaigns.

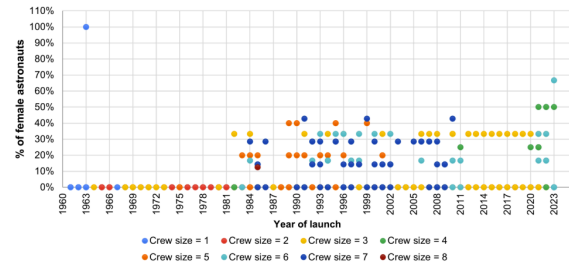


Figure 6-1: Female participation per mission over time [6-1].

6.2 Technological Advancements and Their Role in Selection

Several key technologies have had a concrete impact on astronaut selections, shaping the qualifications and profiles sought in candidates. Here's a look at some of the most influential technologies:

6.2.1 Advanced simulation and Astronauts' training technologies:

Virtual Reality (VR) and Augmented Reality (AR): These technologies are used for immersive training environments, allowing candidates to practise complex scenarios and spacecraft operations in simulated conditions. This has shifted the emphasis towards candidates who can excel in virtual environments and adapt quickly to new technology.

High-fidelity simulators: Simulators that replicate spacecraft controls, environmental conditions, and mission scenarios have become more sophisticated. This allows for more rigorous assessment of a candidate's technical skills and problem-solving abilities.

6.2.2 Medical and health monitoring technologies:

Biometric monitoring: Technologies that track physiological parameters such as heart rate, stress levels, and sleep patterns provide valuable data on how candidates handle stress and physical strain. This has led to a greater focus on candidates' overall health and psychological resilience.

Advanced medical diagnostics: The development of better diagnostic tools and health monitoring systems has improved the ability to assess candidates' physical and mental health, ensuring they can withstand the rigours of space travel.

Psychological assessment tools: Enhanced psychological testing and assessment tools help evaluate candidates' mental and emotional stability. This includes tools to assess stress response, teamwork dynamics, and resilience.

Telemedicine and remote support: Technologies that enable remote medical support and consultation are important for ensuring astronauts can handle medical issues in space. Candidates are now evaluated on their ability to interact with these systems and manage health issues autonomously.

6.2.3 Engineering technologies and processes improvement

Automated systems and robotics: Modern spacecraft are equipped with highly automated systems and robotic technologies. Candidates now need to demonstrate proficiency with these systems, including the ability to troubleshoot and interact with advanced automation tools.

New capsules and private spacecraft: The introduction of spacecraft like SpaceX's Crew Dragon has created new criteria for astronaut selection, focusing on familiarity with modern spacecraft systems and operations.

High-Bandwidth communications systems: The ability to handle high-data-rate communication systems is crucial for modern astronauts. This includes managing data from various sensors and communication networks. Candidates are now evaluated on their ability to work with these advanced communication tools.

Advanced data analysis tools: The use of sophisticated software for data analysis and mission planning requires astronauts to have strong analytical skills and experience with data-driven decision-making. Some of the ISS research can be found on the ISS research NASA page [2].

Space science instruments: The increasing focus on scientific research in space has led to the use of specialised scientific instruments and experiments. Candidates with expertise in fields such as biology, chemistry, and physics are now sought to conduct and manage these experiments.

Microgravity research tools: Technologies for conducting experiments in microgravity environments require astronauts to have a solid understanding of how to use and interpret results from these tools.

Life support systems: Advanced life support technologies, including closed-loop systems for air, water, and waste management, demand that astronauts have a deep understanding of these systems. Selection processes now include assessments of candidates' ability to manage and maintain these systems.

5. Discussion

The expanded analysis presented in this paper offers valuable insights into the evolving landscape of astronaut selection. By examining regional trends, skill set evolution, and the distinction between public and private astronaut profiles, we provide a broader understanding of how societal and technological changes have reshaped astronaut selection processes.

One of the key findings is the increasing regional diversity among astronaut selections. Historically, astronauts predominantly came from a small number of spacefaring nations such as the United States and Russia. However, our analysis shows that in recent years, space agencies have expanded their astronaut selection to include candidates from countries with emerging space programs. This trend reflects a growing emphasis on international collaboration in space exploration, driven by both governmental and private entities. The involvement of new space players, particularly from Latin America, Africa, and Southeast Asia, highlights the democratization of access to space.

The evolution of skill sets required for astronauts has also shifted significantly. While early astronaut profiles emphasized military backgrounds and piloting skills, contemporary astronauts are increasingly selected for their scientific expertise, adaptability, and experience with complex technologies such as robotics, data analysis, and virtual reality simulations. This shift reflects the changing nature of space missions, which are now more focused on long-term scientific research, exploration, and collaboration in space environments like the International Space Station (ISS).

Moreover, our analysis indicates that public and private astronaut profiles are beginning to diverge. Private space companies, particularly those involved in space tourism and commercial missions, tend to prioritize different skill sets, focusing more on customer service, safety protocols, and communication skills. In contrast, public agencies such as NASA and ESA continue to prioritize candidates with strong scientific and engineering backgrounds for their long-term research and exploration missions.

The societal changes over the past few decades, including the growing emphasis on diversity and inclusion, have also played a significant role in astronaut selection. Initiatives aimed at promoting gender and ethnic diversity have led to a more inclusive pool of candidates, as reflected in the increasing number of female astronauts and astronauts from underrepresented communities. Similarly, advancements in technology, such as the use of virtual reality for training and telemedicine for health monitoring, have

enabled a more diverse range of candidates to successfully meet the physical and mental challenges of space missions.

Finally, the analysis of crew size and mission type underscores the importance of flexibility in crew composition. Smaller crews have traditionally been favored for early exploration missions, while larger, more diverse crews are increasingly being used for long-term scientific research and space station operations. The need for adaptability in crew size and composition will only increase as humanity embarks on more ambitious missions, such as returning to the Moon and exploring Mars.

6. Conclusions

This study provides a comprehensive analysis of the evolution of astronaut profiles, emphasizing the growing importance of diversity, adaptability, and scientific expertise in modern astronaut selection. Our findings highlight several key trends, including the increasing regional diversity of astronaut selections, the shift in skill sets required for space missions, and the growing differentiation between public and private astronaut profiles.

The research also underscores the impact of societal changes and technological advancements on astronaut selection criteria. The increasing emphasis on diversity and inclusion has broadened the pool of candidates, allowing space agencies and private companies to select astronauts from a wider range of backgrounds and experiences. Similarly, technological advancements in training, health monitoring, and spacecraft operations have enabled astronauts to meet the ever-increasing demands of space missions.

As space exploration continues to evolve, future astronaut selection processes will need to adapt to the challenges posed by long-duration missions, increased international collaboration, and the growing role of private space companies. Our study provides a foundation for understanding these challenges and offers insights that can help shape more inclusive and effective selection processes.

Future research should continue to explore the intersection of societal changes, technological advancements, and astronaut selection. Additionally, more qualitative data, including interviews with current and former astronauts, could provide valuable insights into how these factors are experienced at a personal level and how they shape the success of space missions.

Acknowledgements

We extend our deepest gratitude to the Space Generation Advisory Council (SGAC) for its unwavering support and for providing the platform for our initiative, DIVINAS (Diversity IN Astronaut Selection), within the Diversity and Gender Equality Project. As members of this esteemed organization, we are privileged to conduct impactful research in astronaut selection and diversity in the space sector. SGAC's continued encouragement and steadfast support have been instrumental in enabling us to advance our objectives. We take great pride in being part of an organization that fosters inclusivity and drives progress in the field of astronautics.

We would also like to express our heartfelt appreciation to astronauts Nicole Stott and Victor Hespánha, whose interviews provided invaluable insights and helped deepen our understanding of key issues in the field.

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